

HealthConnect Solution Development & Implementation

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Analyst: Poloko Dignity Ranna

Dataset: HealthConnect Appointment Dataset

Part 1: Review of Week 4 Foundation

I defined the primary business problem as HealthConnect Clinic's operational challenges stemming from high patient no-show rates. The secondary problem was their difficulty in understanding the specific factors associated with these missed appointments, making it impossible to implement targeted solutions.

Proposed Week 4 Approach:

My approach was a structured 4-phase plan:

- **Data Understanding & Quality Assessment:** Profiling the dataset to identify missing values, data types, and inconsistencies.
- **Exploratory Data Analysis (EDA):** Formulating and exploring specific business questions to understand patterns in patient behavior and appointment characteristics.
- **KPI Development:** Proposing key performance indicators (KPIs) to measure progress and support decision-making.
- **Dashboard Development:** Planning to create a Power BI dashboard to visualize findings and KPIs.

Key assumptions or limitations identified:

Assumptions

- The "none" value in reminder channel correctly indicates that no reminder was sent, rather than entry error
- The calculated fields age_group and booking_lead_days are accurate and derived consistently

Limitations

- The dataset is entirely fictional and may not reflect real-world healthcare data patterns
- No information about appointment rescheduling is available, which could impact the interpretation of cancellation data

Proposed Approach for Week 5

I planned to move from planning to implementation. This included:

- **Data Preparation:** Handling missing values, converting dates, and validating calculated fields.
- **Exploratory Data Analysis:** Creating summary statistics & visualizations to identify no-show patterns.
- **Statistical analysis:** Outline hypothesis & identify variables to be tested
- **Dashboard Development:** Building KPI cards, designing layout, and creating interactive visuals.

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Reporting: Summarizing findings and providing recommendations.

Part 2: Track-Specific Assignment (Data Analytics)

Data Preparation

Prepare the HealthConnect_Appointment_Data.csv for reliable analysis by addressing data quality issues identified in Week 4 using pandas and provide a cleaned dataset to be used for analysis.

Missing Values

- **Distance_to_clinic_km:** replaced with median because assigning 0 meant that clients practically stayed in the clinic which is not logical
- **Waiting_time_minutes:** checked for appointment outcome before assigning values, assigned median to clients who showed up for the appointment and 0 to all clients who cancelled/missed the appointment
- **Reminder_sent:** replaced 'none' values with 'no reminder' to make it clear that it meant not sent rather than 'nan'

Data types: changed Booking_date & appointment_date data type from string to datetime

Data Validation: recalculated booking_lead_days using the formatted dates to make sure there's no human error.

Duplicates: check for duplicates indicated 0 duplicates

Data categorization: categorized distance_to_clinic_km into distance_group to simplify analysis and visualization

Exploratory Data Analysis (EDA)

Investigate appointment attendance and no-show patterns by analyzing the relationships between the target variable (appointment_outcome) and various attributes as outlined

Numerical Variable vs No-Show Analysis

- **Age:** No-Show patients have a slightly lower average age (48.2) compared to Attended (49.2) and Cancelled (50.1).
- **Distance:** No-Show patients have the highest average distance (10.52 km), followed by Cancelled (10.11 km), and Attended (9.67 km).
- **Waiting_time:** No-Show and Attended patients have similar average waiting times (23.8 vs. 24.3 minutes).
- **Booking_lead:** No-Show patients have the highest average lead time (34.5 days), compared to Attended (24.5 days) and Cancelled (29.6 days).
- **Prev_no_show:** Patients with a history of No-Shows have the highest average (0.64), compared to Attended (0.46) and Cancelled (0.42).

Key Visualizations and Interpretations (from Python):

No-Show Rate by Age Group & Gender (grouped bar chart):

- **Observation:** Females aged 45-54 and Males aged 18-24 have the highest no-show rates at 52.5% and 56.0% respectively. Ladies show increase in no-show rate between and decrease when they reach 65+ while man on the other hand shows slight decline in no-show rate as they grow older.
- **Insight:** This suggests demographic-specific barriers, such as work conflicts, childcare and the likes, hence we should introduce demographic specific interventions.

No-Show Rate by Appointment Day & Time (grouped bar chart):

- **Observation:** The no-show rate is high on Sundays and Fridays. Evening appointments have the highest no-show rate across most days
- **Insight:** This points to a "weekend effect." Patients might be more likely to forget appointments scheduled at the end of the week, or they may be more likely to schedule conflicting plans on weekends.

Appointment Outcome by Previous No-show (stacked bar chart):

- **Observation:** The percentage of no-shows increases dramatically with the number of previous no-shows
- **Insight:** Patients with a history of no-shows are a high-priority group for intervention.

Reminder & Logistics No-show Rate (stacked bar chart):

- **Observation:** No-show rate increases with distance up to 40km when a reminder is sent, with a sharp spike at 30-40km (77%), then drops at 40km+.
- **Insight:** Patients living 30-40km away who receive reminders have a 77% no-show rate which is 24% higher than those without reminders, suggesting logistical barriers (transport, traffic, parking) outweigh the effect of reminders.

Monthly Trends (combined line & bar chart):

- **Observation:** May has the highest number of bookings but the lowest turn-up rate.
- **Insight:** The clinic should investigate why May has such a high no-show rate. Is it due to seasonal factors, holidays, or other external events?

Reminder Channel Efficiency (combined line & bar chart):

- **Observation:** Email is the most used channel but has a no-show rate of **48%**, making it the second least effective channel after WhatsApp.
- **Insight:** The clinic should consider shifting more resources toward SMS reminders, especially for high-risk patient segments.

No show rate by Appointment type (line chart):

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Observation: Follow-up appointments have the highest no-show rate, while Diagnostic Tests have the lowest. This suggests that patients may perceive follow-ups as less urgent or may forget to attend after their initial visit

- **Insight:** consider offering virtual follow-up consultations to reduce attendance barriers.

Statistical Testing

Hypothesis and variables to be tested

- Younger demographics are more likely to no-show (age_group, appointment_outcome)
- Distance to the clinic influences no show (distance_to_clinic_km, appointment_outcome)
- Waiting time has influence on future appointment outcome (waiting_time_minutes, appointment_outcome)
- Patients who book far in advance are most likely to miss the appointment(booking_lead_days, appointment_outcome)
- History of no-show is a strong predictor of future no-show behavior (previous_no_shows, appointment_outcome)

Dashboard Development

KPI Cards

In the previous week we outlined, defined and identified business questions associated with our KPI cards, for this week we build, denote formulas and interpret the cards

Total Bookings: created using COUNT(appointment_Id) and we got 5000 which imply that the clinic handled 5,000 appointments during the reporting period. This establishes the scale of clinic operations. When filtered by month, this KPI helps identify peak booking periods.

Total Attended: Computed using CALCULATE(COUNT(cleaned_data[appointment_Id]), cleaned_data[appointment_Id] = 'Attended') which gave 2314. This shows that only 2,314 out of 5,000 appointments were attended. This means ~41% of appointments were either no-shows or cancellations. When filtered by demographics, this KPI reveals which patient groups are most likely to attend.

Total No-show: we used CALCULATE(COUNT(cleaned_data[appointment_Id]), cleaned_data[appointment_Id] = 'No-show) and got 2,423 which mean 2,423 appointments resulted in no-shows. This represents a significant loss of revenue and operational inefficiency. Analysis shows that appointment type (e.g., Follow-up), day (Sundays, Fridays), and time (Evenings) have higher no-show rates.

Show rate: DIVIDE(Total Attended], [Total Bookings], 0) resulting in 46.28%, which implies that only 46.28% of all booked appointments are attended. This means the clinic loses more than half of its appointment slots to no-shows and cancellations

No-Show Rate: Computed using DIVIDE([Total No-Shows], [Total Bookings], 0) which gave 48.46% this mean that nearly 1 in 2 appointments (48.46%) result in no-shows. This is alarmingly high.

Previous no show: COUNTROWS(FILTER(VALUES('cleaned_data'[Patient_Id]),CALCULATE(SUM('cleaned_data'[previous_no_shows]) > 0))) was used and showed 1232 this translates to approximately 1,000 unique patients have missed at least one appointment in the past. This segment is at high risk of future no-shows.

Dashboard or Analytical Visualization

We used cleaned dataset.csv to model our data in Power BI, this week we primarily focused on developing the executive dashboard that provides high-level, at a glance view of clinic performance for management and decision makers. It included;

Top Bar: KPI Cards for Total Bookings, No-Show Rate, Show Rate, and Total Attended.

Middle Section (Left): A donut chart showing the breakdown of Appointment Outcome (Attended, NoShow, Cancelled),

Middle Section (Middle): A column chart showing No-Show Rate by Day of Week to identify problematic days.

Middle section (Right): Combo chart showing reminder channel by appointment type & reminder sent

Bottom Section (left): A line chart showing Monthly Trends in Bookings and No-Shows to identify seasonal patterns.

Bottom Section (right): horizontal column chart showing no show rate by appointment type

Side bar: Slicers for Gender, Appointment time, Appointment Type, and Month are available on the left side for interactive filtering.

Business Insights Reporting

We now translate the findings from the EDA and KPI analysis into actionable recommendations for HealthConnect Clinic.

Findings

- Nearly 1 in 2 appointments (48.46%) result in no-shows. This represents a significant loss of revenue and operational inefficiency. The clinic is losing almost as many appointments as it is keeping
- While Email is the most used reminder channel, SMS achieves the highest attendance conversion
- Sunday and Friday have the highest no-show rates. Evening appointments show elevated no-show rates across most days.
- Patients may perceive follow-ups as less urgent or may forget to attend after their initial visit. Diagnostic Tests have the lowest no-show rate.
- Approximately 1,232 unique patients have missed at least one appointment in the past. Patients with a history of no-shows are at significantly higher risk of future no-shows.

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Patients living 30-40km away who receive reminders have a 77% no-show rate which is 24 percentage points higher than those without reminders. This suggests logistical barriers (transport, traffic, parking) outweigh the effect of reminders.

Solutions

- Prioritize SMS reminders as the primary intervention channel to reduce this rate because it has proven to be the most effective channel
- Shift communication budget from email to SMS reminders, especially for high-risk patient segments.
- Implement enhanced reminder protocols (e.g., secondary SMS or call) for Sunday and Monday appointments.
- Offering virtual follow-up consultations to reduce attendance barriers
- Flag high-risk patients for proactive outreach (e.g., morning-of appointment check-ins or phone calls 48 hours in advance)
- Offer virtual consultations or transportation assistance for patients living >30km away.

Part 3: Assumptions, Limitations, Risks & Dependencies

This section reviews the Week 4 findings and updates them based on Week 5 work, i.e. identifying resolved ones and ones yet to be worked out.

Assumptions

Relevant: The "none" value in reminder_channel correctly indicates no reminder was sent.

We categorized 'none' as a distinct 'No Reminder' category in the dataset and dashboard.

Resolved: The calculated fields (age_group, booking_lead_days) are accurate and derived consistently.

We recalculated and validated booking_lead_days to ensure their accuracy, fixing any potential errors.

Limitations

Relevant

The dataset is fictional and may not reflect real-world healthcare data patterns. Making insights directional and illustrate the process.

No information about appointment rescheduling is available, we cannot differentiate between a patient who cancels and reschedules from one who simply cancels. This impacts the interpretation of 'cancellation' data. A new data capture point for rescheduled appointments is recommended for future projects.

Risks

Resolved: Inconsistent handling of missing values could lead to misleading conclusions. To mitigate this risk, we used a consistent and documented strategy (median imputation for numerical values) **Relevant:**

It is unclear if `waiting_time_minutes` is an actual historical wait time or an estimate. Some values are missing, this creates uncertainty in our analysis. If it's an estimate, it's less reliable. If missing data is not random, our analysis could be biased. We assumed it's an actual historical wait time for the analysis and replaced the missing values with median

Cross-Track Collaboration Dependency

Interacted with Project Management Intern(Project leader for pod 007)

What was expected: Shared my Week 5 deliverables including key findings (48.46% no-show rate, 77% no-show rate for 30-40km group, SMS as most effective channel). Discussed the 6 KPIs I developed and their business relevance.

Why it was relevant: The PM needed to understand which KPIs are most critical for tracking intervention success and project progress. This helps her document success metrics in the project plan and coordinate dependencies across tracks.

What changed: Through our discussion, I now understand that No-Show Rate and Show Rate are the primary KPIs that should be used as starting points for post-intervention comparison

Interacted with Data science Intern(Dineo Nadine Sithole)

What was exchanged: Shared my EDA key findings, 6 KPI cards (Total Bookings: 5,000, Total Attended: 2,314, Total No-Show: 2,423, Show Rate: 46.28%, No-Show Rate: 48.46%, Previous No-Show Patients: 1,232), and project reports. She shared her Baseline Modelling Report, including her feature engineering approach (`previous_noshow_rate`, `had_previous_noshow`, `reminder_and_channel`) and modeling results.

Why it was relevant: Our tracks need to align with insights and feature definitions, from our independently built analyses our dataset understanding was consistent, cross-validated our core numbers (5,000, 2,314, 2,423)

What changed:

- Identified a scope difference in our reminder channel analysis, which made me cross-check my EDA report where I found that our results are actually the same, it was an observation error from my side. This strengthened confidence in the reminder-channel, finding as a genuine, reproducible pattern.
- I now understand the role that `waiting_time_minutes` plays, it's not necessary for predictive analysis since it's only recorded after the patient arrives, but it is useful for historical analysis. I will keep it in my dashboard with a clear disclaimer.
- Based on her modeling approach, I will add `previous_noshow_rate` to my dashboard metrics in Week 6, as it is more informative than the raw count for risk assessment.
- She will add distance × reminder interaction and age/gender patterns as features in her Week 6 model, based on my EDA findings.

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77% no-show rate for 30-40km group with reminders, SMS as most effective channel (48%), demographic patterns (females 45-54 and males 18-24 highest risk) and . She shared her baseline modeling approach and feature engineering decisions.

Our tracks need to align with insights and feature definitions. Her model features should reflect the strongest drivers of no-show behavior identified in my EDA

The following changes will be adopted ○ She will add distance × reminder interaction and age/gender patterns as features in her

- Week 6 model ○ She confirmed waiting_time_minutes should be dropped from predictive models as it's only recorded after the patient arrives. But I will keep waiting_time_minutes in my dashboard for historical insight only with a clear disclaimer
- We confirmed our insights are aligned, validating that we're both working toward the same goal